

Project Proposal: What makes it viral?

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1. Business Value

1.1 Background

Short-form video platforms such as Instagram Reels have become major channels for online content consumption and digital marketing. Millions of short videos compete for user attention every day, but only a small portion achieve viral reach. Understanding which measurable content characteristics are associated with higher engagement has become increasingly important for content creators, marketers, and platform analysts.

1.2 Problem Definition & Value Proposition

This project investigates which content-related factors are associated with viral performance on Instagram Reels. Specifically, it aims to examine whether measurable features such as hashtags, content category, and engagement indicators can help explain why some reels achieve substantially higher reach than others.

The expected value of this analysis is to provide empirical insight into how short-form content spreads and to identify which observable variables are most strongly associated with viral engagement.

2. Available Data

2.1 Data Acquisition

The project will use the What Makes an Instagram Reel Go Viral dataset obtained from Kaggle.

The dataset contains reel-level performance records collected from Instagram short-form content. It includes both engagement-related metrics and content metadata, making it suitable for exploratory analysis and predictive modeling.

2.2 Data Review

The dataset contains variables related to reel performance and content characteristics. These include engagement indicators such as:

- views
- likes
- comments
- shares

It also includes descriptive content attributes such as:

- hashtag-related information
- content category or niche

- metadata describing the reel

These variables allow analysis at the content level rather than the individual user level.

3. Formulation

3.1 Choice of Algorithms and Rationale

The project will begin with exploratory data analysis to understand the distributions of engagement metrics and the relationships between content features and reel performance.

For predictive modeling, Logistic Regression will be used as a baseline model because it is interpretable and suitable for binary classification tasks.

In addition, Random Forest will be applied because it can capture nonlinear interactions among variables and provide feature importance measures that help interpret which content characteristics are more influential.

3.2 Input & Output Variables

Input variables

- Creator audience size
- Reel length
- Hook effectiveness
- Content quality
- Retention rate
- Shares and reach

Output variable

- virality score

A reel will be classified as viral based on engagement-related thresholds, such as top-percentile performance in views or combined engagement metrics.

3.3 Expected Challenges and Alternatives

One challenge is that the definition of “viral” may affect the modeling results. Different threshold choices may lead to different class distributions.

Another limitation is that engagement is partly influenced by platform recommendation algorithms and audience-specific factors that may not be directly observable in the dataset.

If classification performance is limited, the project may place greater emphasis on exploratory analysis and descriptive interpretation of engagement patterns rather than predictive accuracy.